

Ram Sankar Koripalli

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SUMMARY

Software engineer with 5+ years of experience building distributed systems, backend microservices, and ML infrastructure at scale. Tech lead for telemetry on 100M+ Amazon devices; previously shipped Go cloud-storage services for 100+ enterprise customers at HPE. Hands-on with LLM inference, GPU/CUDA kernel optimization, and high-throughput, low-latency systems. Seeking AI infrastructure, distributed systems, and backend engineering roles.

TECHNICAL SKILLS

Languages Python, Go, C++, C, Java, JavaScript, SQL, Bash, CUDA

AI / ML PyTorch, TensorFlow, ONNX, LangChain, LLM inference, FP16 quantization, GPU kernels, scikit-learn

Distributed Systems Microservices, gRPC, REST, Kafka, NATS, Redis, event-driven architectures, telemetry & observability

Infra & DevOps Docker, Kubernetes, Helm, Prometheus, Jenkins, GitHub Actions, AWS (EC2, S3, DynamoDB)

Databases PostgreSQL, Oracle, MySQL, DynamoDB, Redis

Web Frameworks Django, Flask, Streamlit, Oracle APEX

PROFESSIONAL EXPERIENCE

Amazon (Lab126) — Software Development Engineer II Sep 2025 – Present
Device OS Metrics & Telemetry Platform Sunnyvale, CA

- Tech lead for the Device OS Metrics platform across FireOS, VegaOS, and AOSP-based platforms; architecting scalable telemetry and metrics-collection systems that power observability for 100M+ active devices.
- Designed a rule-based metrics triggering framework that lets 50+ internal teams declaratively define conditional logic, eliminating manual instrumentation and accelerating time-to-metric from days to hours.
- Built a dynamic metric config registration/deregistration system that honors user consent and multi-user profile switching, ensuring compliance with Google MBA privacy policies across the device fleet.

Hewlett Packard Enterprise — Software Development Engineer Feb 2024 – Aug 2025
Cloud Storage Platform Durham, NC

- Built high-performance Go microservices powering HPE's cloud storage platform for 100+ global enterprise customers, owning secure configuration management, authentication workflows, and gRPC service contracts.
- Engineered a reusable Health Check Manager adopted across 4 team-owned microservices, standardizing Kubernetes liveness and readiness probes and reducing on-call noise from health-related incidents.
- Pioneered AI-driven unit-test generation using prompt engineering with GitHub Copilot, achieving ~95% generated-test accuracy, lifting code coverage by 15%, and cutting manual test-authoring effort by 70%.

Hewlett Packard Enterprise — Software Engineer Intern May 2023 – Aug 2023
Cloud Data Services Org Durham, NC

- Built a CI quality-metrics dashboard aggregating results from 630 GitHub repositories, giving the cloud data services org real-time visibility into test health and build performance.
- Re-architected result extraction from a poll-based to a push-based pipeline using Amazon S3 and GitHub Actions, achieving a 350% throughput improvement.

Oracle — Senior Member of Technical Staff (SDE II)

Oct 2021 – Aug 2022

RDBMS Test Infrastructure

Bengaluru, India

- Built a relationship-mapping system between RDBMS test suites using Oracle PGX (graph DB) and Redis, enabling test deduplication and cutting redundant executions in large regression runs.
- Designed an automated alerting system to track code-coverage regressions across 1,000+ RDBMS modules, driving long-term code-quality improvements at scale.
- Lifted GCC block-level coverage of Oracle GoldenGate from 52% to 77% by overhauling coverage validation, profiling, and test-suite integration in the CI pipeline.

Oracle — Member of Technical Staff (SDE I)

Jul 2019 – Oct 2021

RDBMS Tooling & ML

Bengaluru, India

- Implemented a SQL/OraTst/log linter using ANTLR, Python, and Java that serves ~600 requests/day and is integrated into the code-review workflow.
- Reduced runtime of a test-selection algorithm by 2.3x by identifying CLOB fetch as the bottleneck and re-implementing it as deferred, batched, multithreaded reads via OCCI.
- Trained an RNN-based text-classification model on 4,500 RDBMS bug reports for automated root-cause identification, accelerating bug triage.

JPMorgan Chase — Machine Learning Intern

May 2018 – Jul 2018

Fraud Detection Research

Hyderabad, India

- Built RNN/LSTM credit-card fraud-detection models in TensorFlow on 120K transactions; bidirectional LSTM achieved 94% accuracy / 95% precision / 85% recall under 10-fold cross-validation.

SELECTED PROJECTS

CUDA-Accelerated Batched GEMM & MLP Inference

Jun 2025 – Aug 2025

- Wrote a tiled shared-memory CUDA kernel for 512×512 GEMM hitting 120 GFLOPS on a T4 GPU — 4x a CPU BLAS baseline and ~12% of cuBLAS — and an MNIST MLP inference path at 13 ms / 1,000 images.
- Profiled and tuned with Nsight Compute; applied FP16 quantization with <0.5% accuracy loss.

LLaMA-Based LLM Inference Engine

May 2025 – Jun 2025

- Deployed a LLaMA 2-1.3B inference pipeline on AWS EC2 g4dn.large (T4) with FP16 quantization, dynamic batching, and CUDA memory optimizations — 25 ms/token latency while serving 6 concurrent requests.

Two Truths One Lie — Generative AI Trivia Game

Jan 2025 – Feb 2025

- Interactive trivia app built with LangChain and Streamlit; players spot the LLM-generated lie among three statements.

Meme Recommendation System

Apr 2023 – May 2023

- Built a Django + DynamoDB recommender that suggests memes from user preferences and engagement; cached embeddings in Redis for a 5x retrieval speedup.

EDUCATION

Texas A&M University — M.S., Computer Science (GPA 4.0/4.0)

Aug 2022 – Dec 2023

College Station, TX · \$1,000 Department of CSE Academic Scholarship (2023)

NIT Tiruchirappalli — B.Tech, Computer Science (CGPA 8.54/10.0)

Jul 2015 – May 2019

Minor in Management Studies · ET Campus Stars Challenger Award (top 34–70 of 5,000, 2018)